

BETTER DAYS HACKATHON – CLICKHOUSE SF

# WORD@LAST

**The Voice Arcade Game Turning 4th-Grade Reading  
Hesitation into Millisecond Telemetry & Immediate Mastery**



**INSERT COIN  
TO START**

## Villain Profile: The 4th-Grade Reading Slump

# 70%

HEALTH BAR



US 4th Graders  
Below Proficiency

### The Attack (The Critical Reading Shift)

In 4th grade, children transition from learning to read to reading to learn. When students stumble over multisyllabic phonics rules, hesitation compounds into silent classroom disengagement.



### Ineffective Weapon (Passive Flashcards)

Fails because it lacks instant feedback and cannot capture where or why a child hesitates.

# The Sub-Vocal Bottleneck

## The Cognitive Gap

When a 3rd or 4th grader labors over words like 'wrestle' or 'knight', their working memory fills entirely with phonetic decoding, leaving zero zero capacity for comprehension.

Working Memory Capacity

**BOTTLENECK  
DETECTED**

Sentence  
Comprehension

95%

Decoding  
Effort  
(sounding out  
w-r-e-s-t-l-e)

Struggling  
Reader

Sentence  
Comprehension

Dual-Modality Processing  
(Visual + Auditory-Motor)

WordBlast  
Player

## The Solution: Orthographic Mapping

Speaking the scrolling word aloud engages dual-modality processing. This repeated visual-to-visual-to-vocal translation builds automaticity—permanently connecting visual spelling to spoken sound.

# UPGRADING FROM ANALOG TO ARCADE MODE



**PLAYER 1**  
(OLD SCHOOL)



PASSIVE FLASHCARDS

DELAYED TEACHER GRADING

BINARY (RIGHT/WRONG)

STATIC DECKS



**PLAYER 2**  
(ARCADE MODE)



REAL-TIME VOICE CONTROLLER



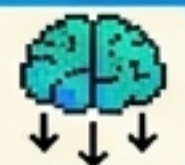
INSTANT VISUAL EXPLOSIONS



MILLISECOND CLICKHOUSE  
HESITATION TELEMETRY



LLM-DRIVEN PROCEDURAL  
WORD DROPS



# Choose Your Weapon: The WordBlast Arcade Cabinet



## Voice as Controller

Children speak falling target words aloud into the microphone to blast obstacles in real time.

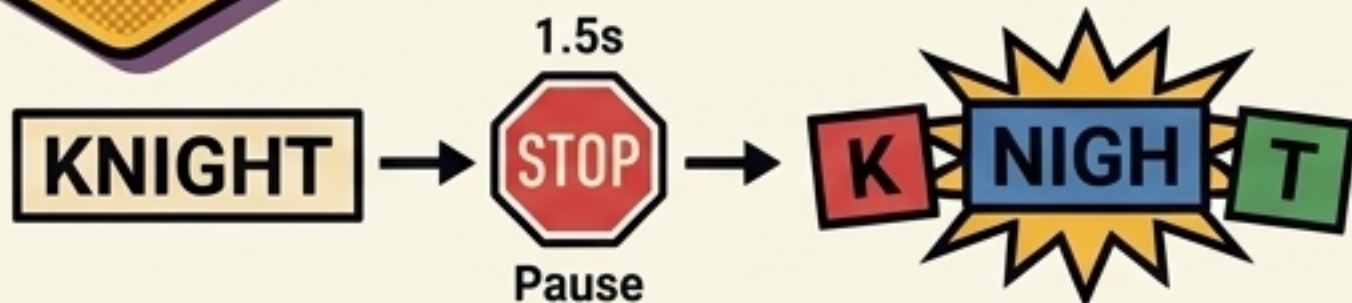
## Instant Visual Hits

Explosive arcade feedback on clean pronunciations keeps engagement and fun high.



## Syllable Scaffolding

Speech pauses over 1.5s automatically trigger split-syllable phoneme highlighting (K-N-I-G-H-T).



## High-Frequency Practice

Converts dry phonics drills into an addicting 10-second arcade gaming loop.

# The Core Gameplay Loop



## Step 1: Word Drop

Phonics obstacle cards fall down synthwave lanes at adaptive speeds.

## Step 2: Voice Blast

Player speaks into the mic. Sub-100ms Web Speech API validates spoken phonemes instantly to score points.

## Step 3: Telemetry Log

Exact hesitation milliseconds and phoneme gaps stream into ClickHouse to shape the next procedural level.

## Dynamic Syllabication

### In-Game Intervention:

When the speech evaluator detects a miss or long hesitation ( $>1,500$  ms), the falling word dynamically splits into color-coded syllable chunks.

1:55

KNIGHT

## Academic Transition

4th grade shifts from single-syllable phonics ('cat') to multisyllabic academic vocabulary. This mechanic explicitly teaches morphological chunking strategies on the fly.

# THE MOTHERBOARD: EXPLODING THE TECH STACK

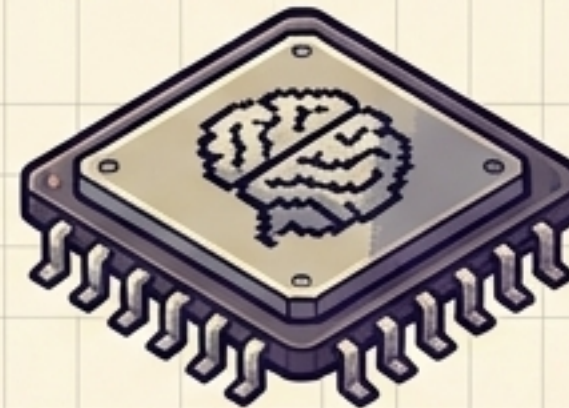
## THE SCREEN/MIC (FRONTEND):

Next.js 14 + Web Speech API.

Handles sub-100ms voice input capture, speech validation & arcade canvas graphics.

## THE GAME DIRECTOR (AI GENERATOR):

LLM JSON Engine. Generates targeted word obstacle drops based on ClickHouse hesitation profiles.



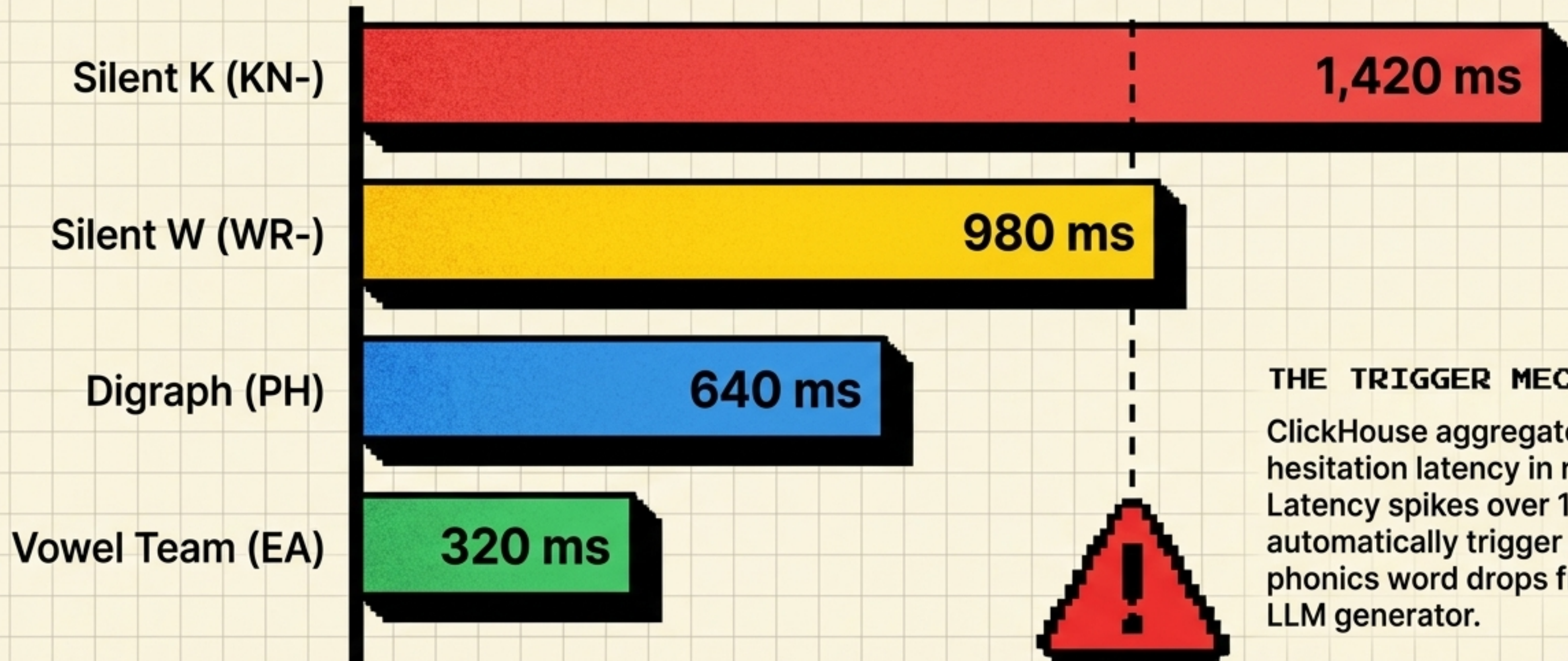
PLAYER 1:  
POSTGRESQL  
(OLTP)

The Transactional State Engine. Manages persistent student accounts, high score leaderboards, and teacher classroom rosters with strict transactional guarantees.

PLAYER 2:  
CLICKHOUSE  
(OLAP)

The Real-Time Telemetry Engine. Ingests millisecond voice hesitation events, computes real-time phoneme error matrices, and powers instant sub-second teacher dashboards.

# THE HESITATION LEADERBOARD: POWERED BY CLICKHOUSE

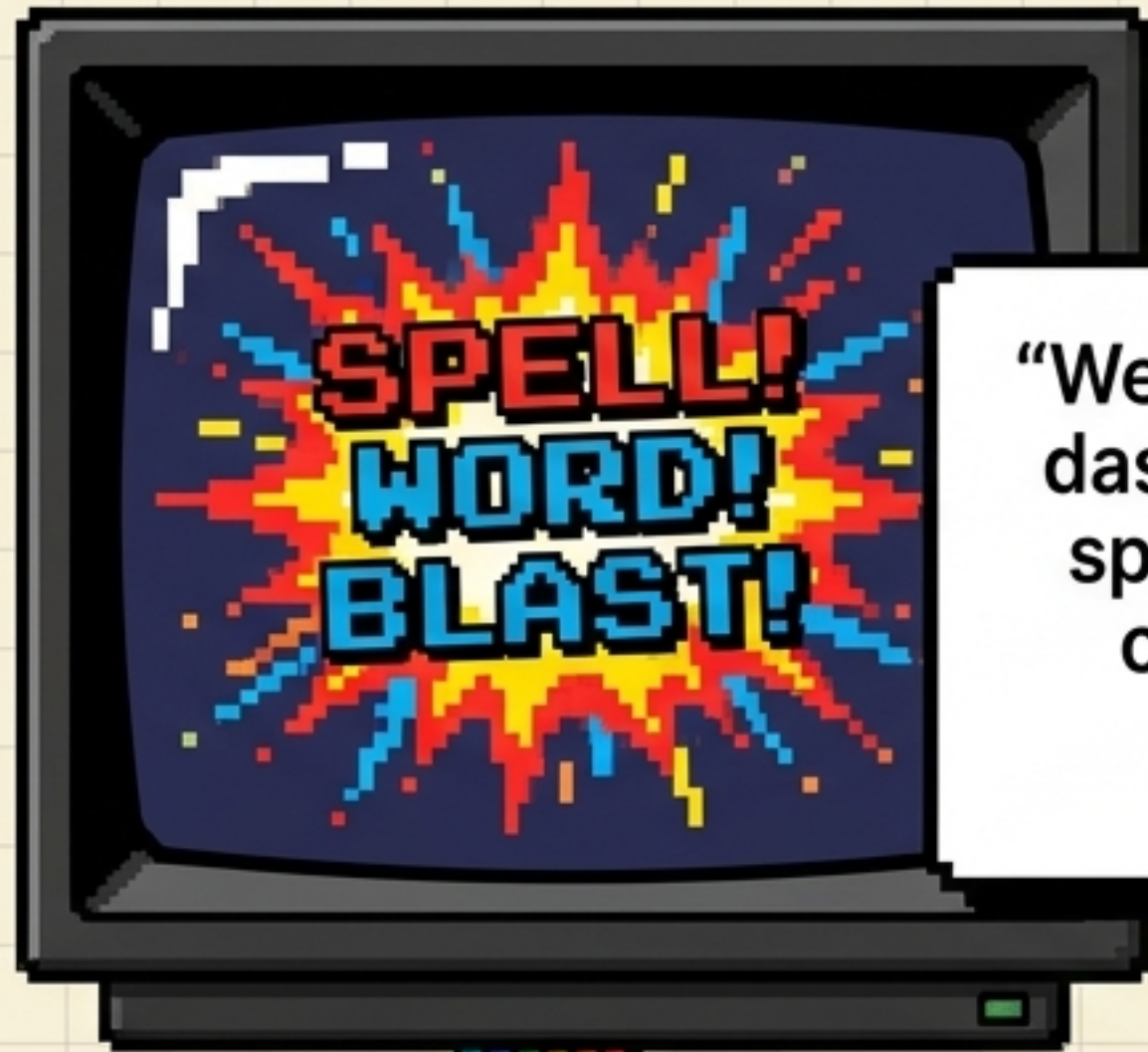


**THE TRIGGER MECHANIC:**  
ClickHouse aggregates speech hesitation latency in real time. Latency spikes over 1,000ms automatically trigger adaptive phonics word drops from the LLM generator.

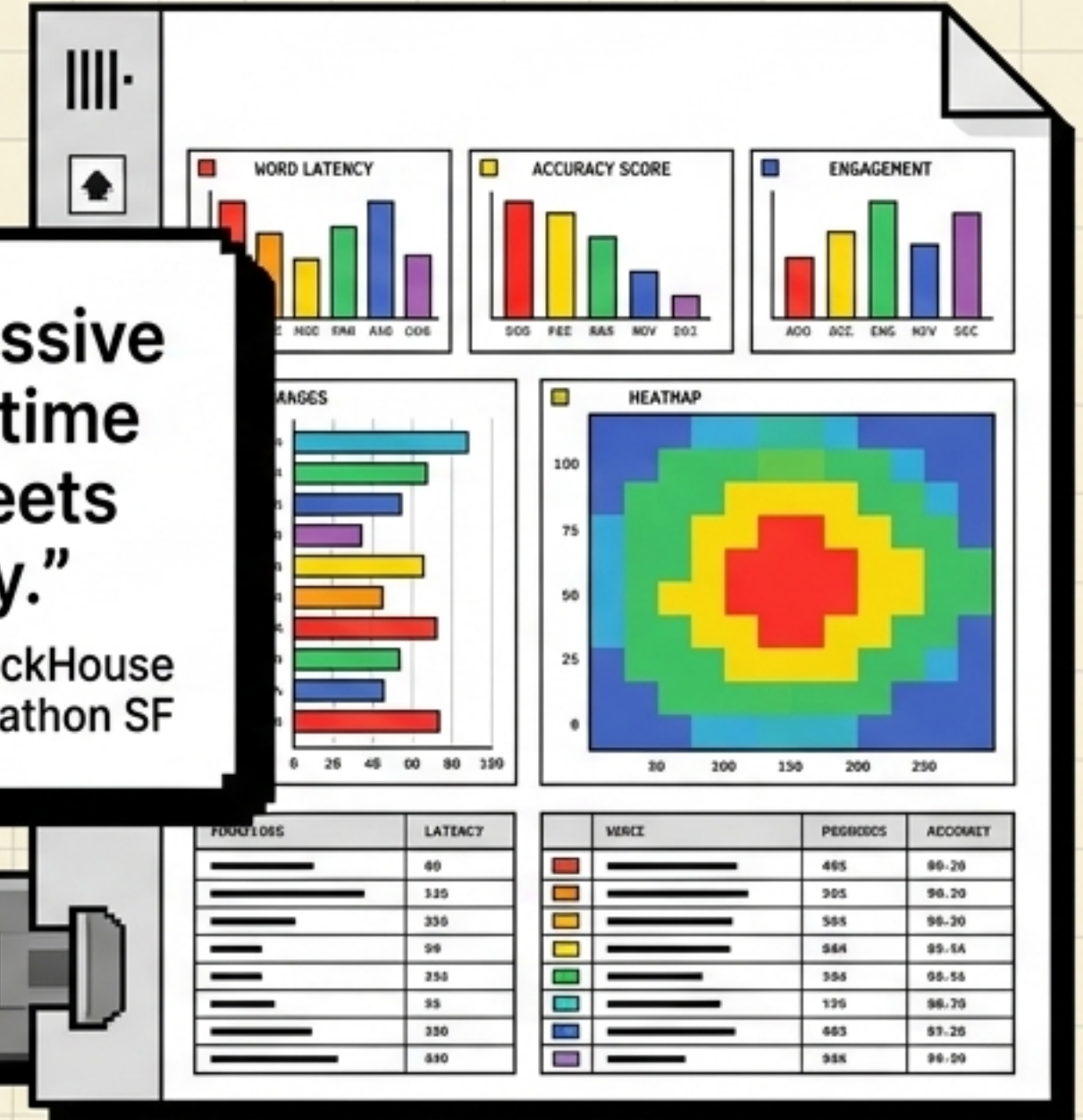
**The Research Reality:** Oral reading fluency (accuracy + speed) is the single strongest predictor of overall reading comprehension. Silent apps cannot measure this.

# THE REAL MAGIC: MEETING CHILDREN WHERE THEY PLAY

CHILD'S VIEW: ARCADE GAME



TEACHER'S VIEW: ANALYTICS DASHBOARD



“We don’t need another passive dashboard. We need real-time speech telemetry that meets children where they play.”

— WordBlast Team | ClickHouse  
“Better Days” Hackathon SF

# BOSS LEVEL: PRECISION REMEDIATION VIA TELEMETRY

Context: Traditional assessments vaguely state: "Reading at a 3.2 level."  
WordBlast reveals the exact structural failure point.

| PATTERN                                | ACCURACY | LATENCY  |
|----------------------------------------|----------|----------|
| Standard Blends (bl-, cr-, dr-)        | 95%      | 450 ms   |
| Digraphs (ph-, ch-, sh-)               | 88%      | 620 ms   |
| Vowel Teams (ea-, oa-, ai-)            | 75%      | 890 ms   |
| Silent-Letter Patterns (kn-, wr-, gn-) | 60%      | 1,850 ms |

## AGGREGATE INSIGHT:

The student decodes 95% of standard blends accurately, but experiences an average latency spike of 1,850 ms exclusively on silent-letter patterns (kn-, wr-, gn-).

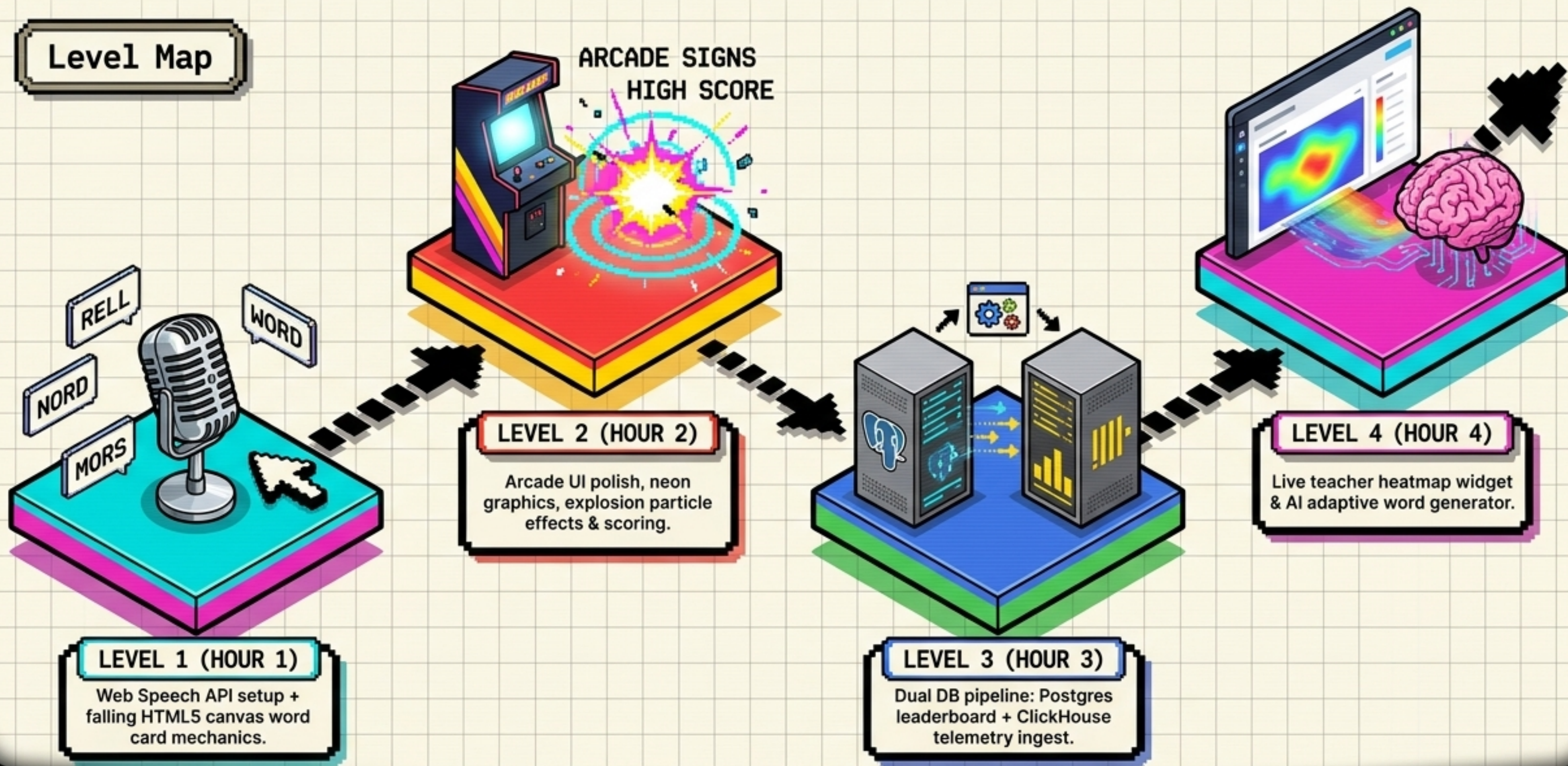
AI  
GENERATION  
ENGINE

## TARGETED FIX:

The AI uses this latency data to dynamically generate a targeted 10-second arcade round, eliminating the exact microbarrier holding the child back.

# THE HACKATHON SPEEDRUN: BUILT IN 4 HOURS

## Level Map



# Game On for Literacy!

Ready to eliminate reading hesitation  
with sub-second data telemetry.